

V-Test - A

Leading third-party testing player to benefit from local AI chip ramp-up;
initiate at OW

China Technology

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Company overview

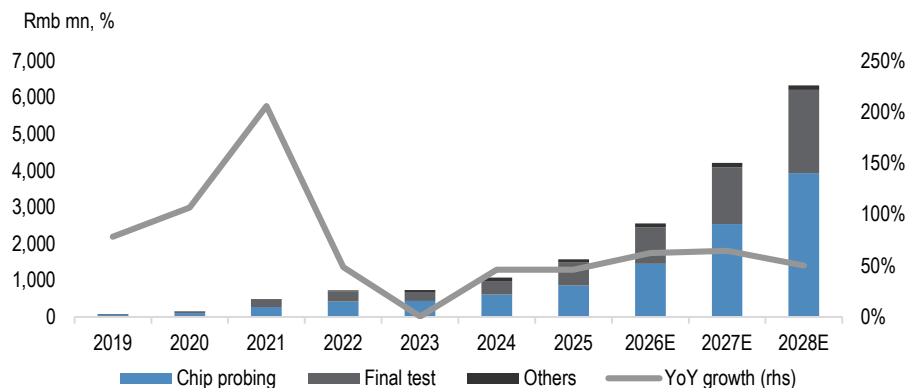
□ Background

- Shanghai V-Test Semiconductor Technology Co., Ltd. (V-Test) is China's leading independent third-party testing service provider. Founded in May 2016, the company listed on the Shanghai Stock Exchange STAR Market in October 2022.
- V-Test generated Rmb1.6bn revenue and Rmb303mn net profit in 2025.
- Current market cap: ~US\$3.0bn; 3M avg. daily volume: ~US\$258mn; SH-HK Stock Connect holdings: 3.36%.

□ Business overview

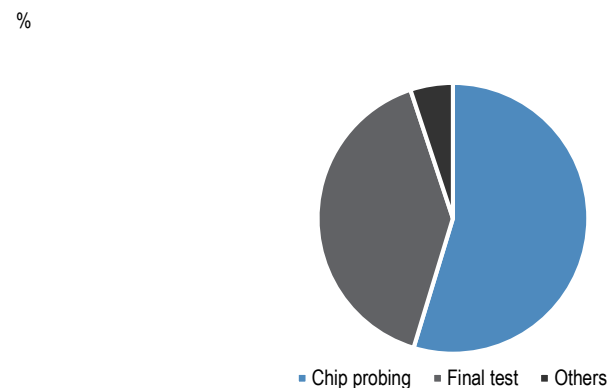
- V-Test specializes in wafer testing (Chip Probing, CP), final test (FT), system-level test (SLT), burn-in, and value-added testing services, providing one-stop testing solutions for chip design houses, foundries and OSATs. The company operates several wholly owned subsidiaries across China's key semiconductor clusters (Wuxi, Nanjing, Shenzhen, Tianjin, and Chengdu).

V-Test's revenue breakdown by segment



Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

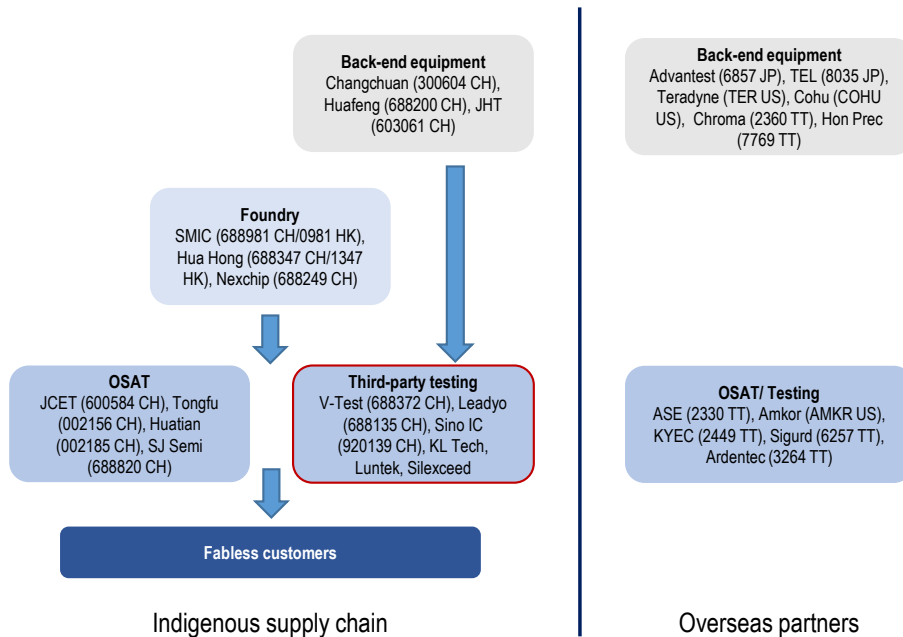
V-Test's revenue mix by segment (2025)



Source: V-Test (2025 annual report).

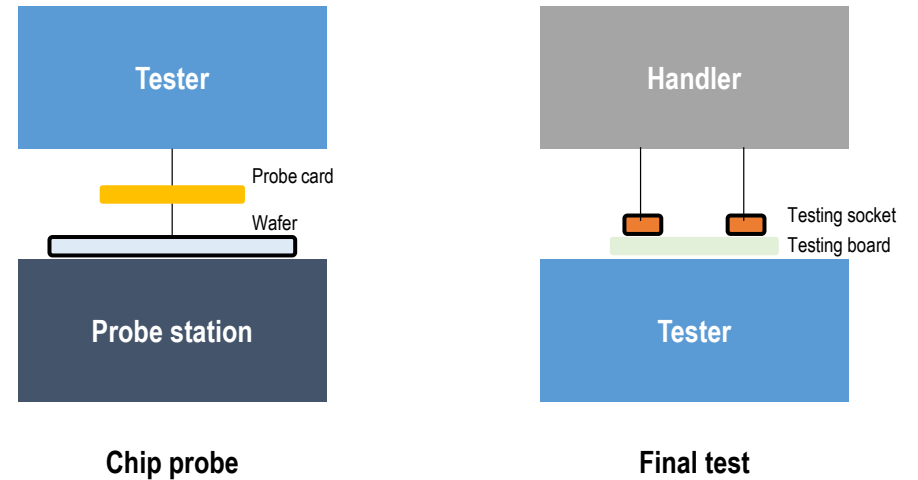
Company overview (cont.)

Key upstream/downstream players in global testing supply chain



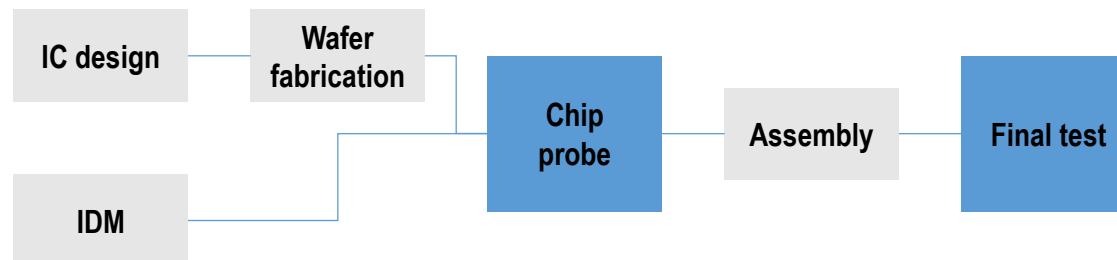
Source: J.P. Morgan.

Illustration of testing process by chip probe and final test



Source: J.P. Morgan.

Testing services within chip manufacturing process



Source: J.P. Morgan.

Investment thesis

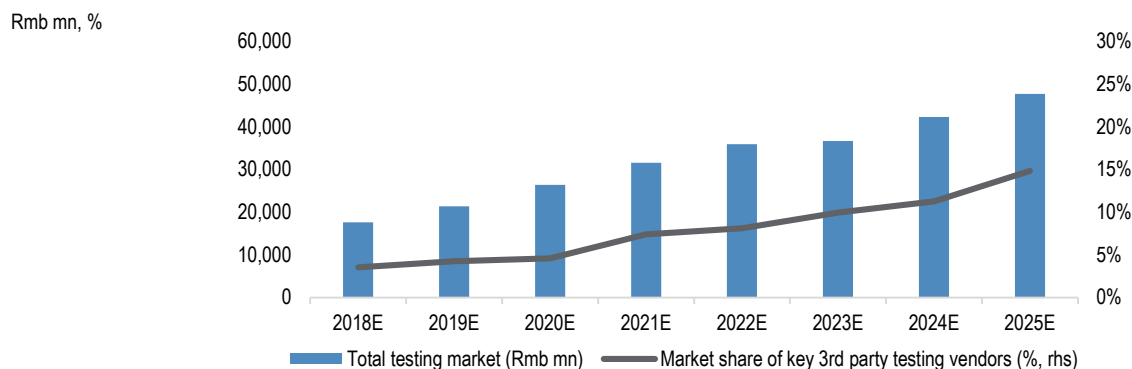
- ❑ **Initiate with OW rating.** V-Test is uniquely positioned to capture the inflection point in the proliferating domestic AI supply chain. We model 59%/73% sales/earnings CAGRs over 2026-28E, with market share doubling from ~10% to 20%+ by 2028E.
- ❑ **Multi-die designs to further amplify AI chip testing demand**
 - ❑ We forecast AI chip testing demand to grow at an 88% CAGR over 2026-28E (vs 23% for the total testing market, ~Rmb89bn in 2028E), reaching around 30% of total testing demand by 2028E.
 - ❑ Due to technology/equipment restrictions, domestic AI chips are forced to accelerate chiplet (or multi-die) architecture adoption, we believe testing vendors will uniquely benefit from this trend, considering: 1) the 1.5-2x multiplier effect on CP volume/complexity and 2) the 1.2-1.5x increase in Final Test (FT) time.
- ❑ **Best positioned to capture upside in AI chip testing demand**
 - ❑ V-Test operates 1,500+ testers (400+ high-end), the largest fleet among A-share pure-play companies. V-Test's counter-cyclical capacity expansion has established a first-mover advantage against customer qualification barriers and high-end equipment lead-time bottlenecks. Given that V-Test doubled its 2025 capex (vs 2024), we believe it has proactively invested to capture upcoming AI chip demand.
 - ❑ We believe V-Test's structurally higher CP exposure (carrying higher gross margin) and premium ASPs for AI chip testing (both CP and FT) driven by equipment exclusivity and engineering intensity could lift margin further. We now expect blended gross margin to expand to 44.6% by 2028E (from 39.4% in 2025).
- ❑ **Initiate at Overweight with Jun-27 PT of Rmb220**
 - ❑ We initiate coverage of V-Test with an Overweight rating and a Jun-27 price target of Rmb220.00 (based on 30x one-year forward P/E).
 - ❑ Our target multiple benchmarks the company's three-year historical average and sits 30% below the trading average of key A-share OSAT peers, reflecting a discount for its relatively limited pricing flexibility in the near term.

Third-party testing – an easily overlooked industry offering exceptional opportunities

□ Domestic substitution and localization trend triggered by tightening restrictions

- We estimate the total market of related testing demand reached Rmb48bn in 2025, achieving ~13% (2021-25) CAGR amid the industry's cyclical trend.
- We believe the domestic substitution wave triggered by decoupling is systematically creating testing demand.
- We estimate the major players (including private ones) in third-party testing accounted for 10-20% market share of China's total testing demand by 2025.
- The combined revenue of listed domestic third-party testing vendors including V-Test, Leadyo and SinoIC was approximately Rmb2.5bn (or ~US\$350mn) in 2025 (according to company reports), delivering a more than 30% five-year CAGR (vs 10%+ by major OSATs).
- Among leading non-listed domestic peers, King Long Technology (KYTEC's former mainland subsidiary) and Xinyun Semiconductor (or Luntek Technology) also have a strong presence in the domestic testing market.

China's total testing market and market share of third-party vendors



Note: including V-Test, Leadyo, SinoIC and Luntek.

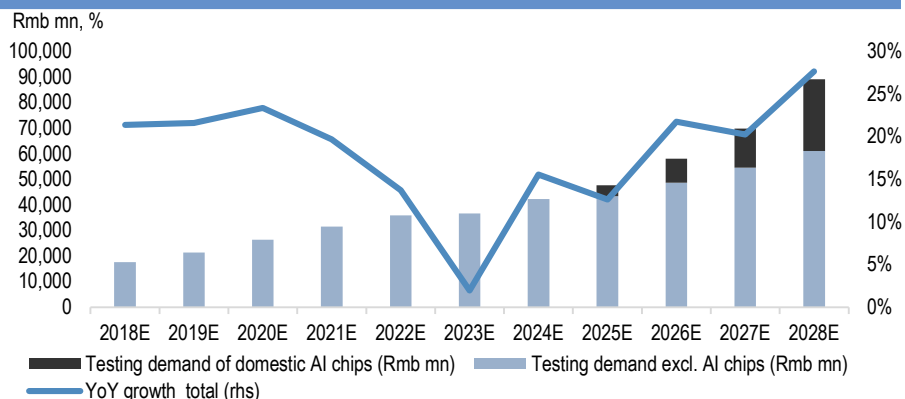
Source: CSIA (23 Jul 2026), V-Test (IPO prospectus), J.P. Morgan estimates.

Domestic AI demand and constraints are nurturing third-party testing industry

□ Incremental growth in market for third-party testing of domestic AI chips

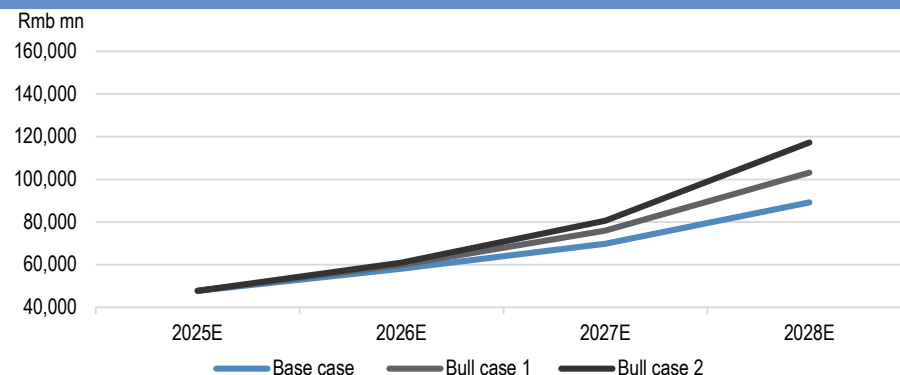
- We expect total testing demand in China to grow at a 23% CAGR during 2026-28E and reach Rmb89bn by 2028E, and AI chip related demand to outpace this at an 88% CAGR, accounting for c30% of total demand by 2028E (from a SD% in 2025). Structural changes include:
 - Highly fragmented nature of domestic AI chip ecosystem amplifying testing demand
 - Testing supply/demand gap widening
 - Full-chain testing demand surging (from CP to FT to SLT)
- Our scenario analysis suggests the adoption of multi-die packages could provide 15-30% upside to the total testing demand.
 - Due to technology/equipment restriction, domestic AI chips are forced to accelerate chiplet architecture adoption – i.e., splitting a large chip into multiple smaller dies.
 - Our assumptions include a two-compute die solution as well as: 1) 1.5-2x multiplier effects on CP volume/ complexity; and 2) 1.2-1.5x testing time for FT.

Total testing demand in China by conventional and AI chips



Source: CSIA (23 Jul 2026), V-Test (IPO prospectus), J.P. Morgan estimates.

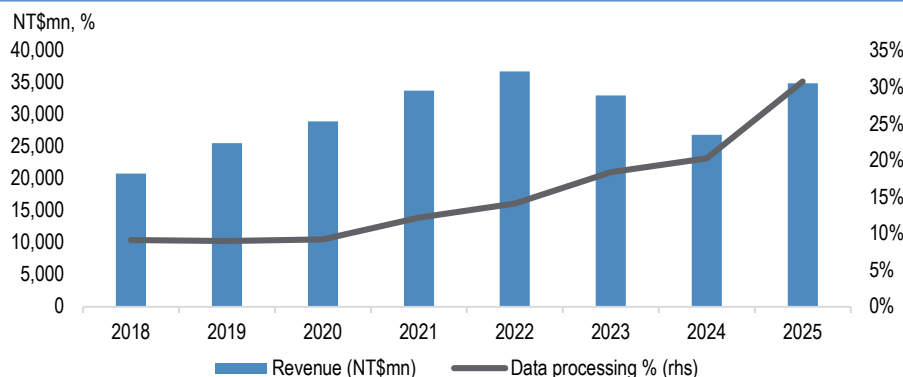
Scenario analysis of total testing demand



Source: J.P. Morgan estimates.

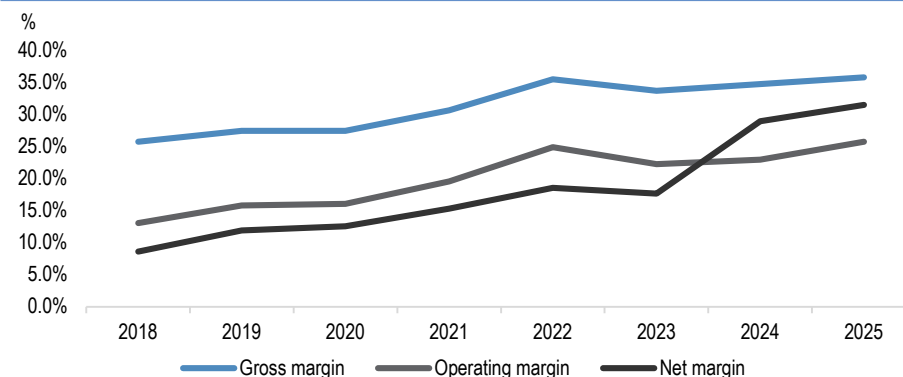
Domestic AI demand and constraints are nurturing third-party testing industry (cont.)

KYEC's historical revenue and contribution by data processing



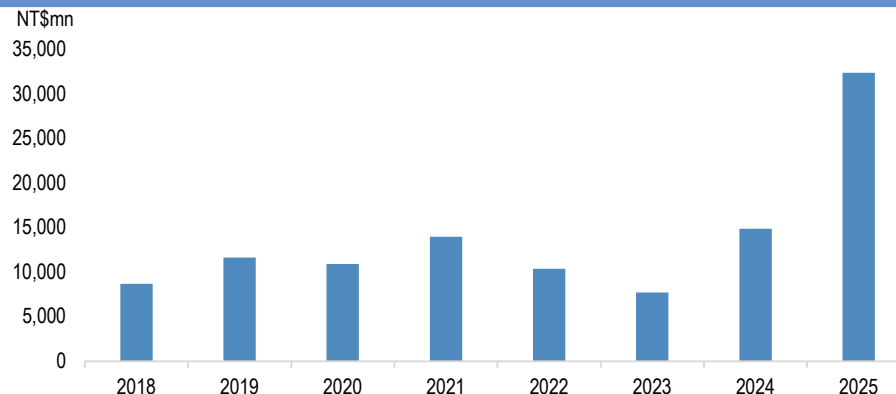
Source: KYEC (Historical public filings, 2018-2025).

KYEC's historical margin trend



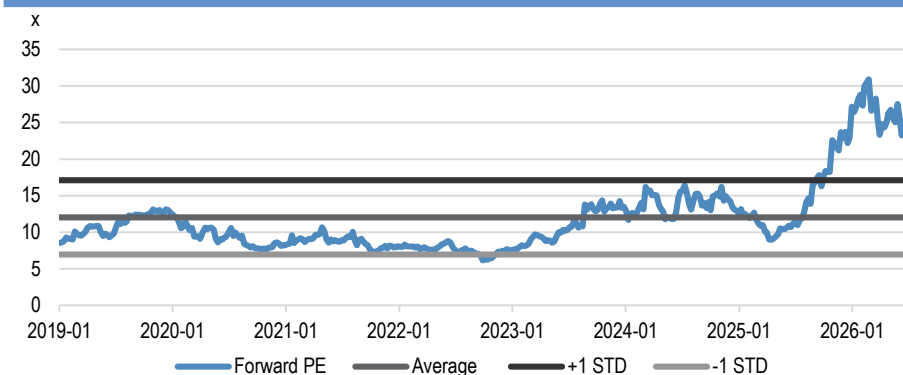
Source: KYEC (Historical annual reports, 2018-2025).

KYEC's historical Capex trend



Source: KYEC (Historical annual reports, 2018-2025).

KYEC's historical PE trading band



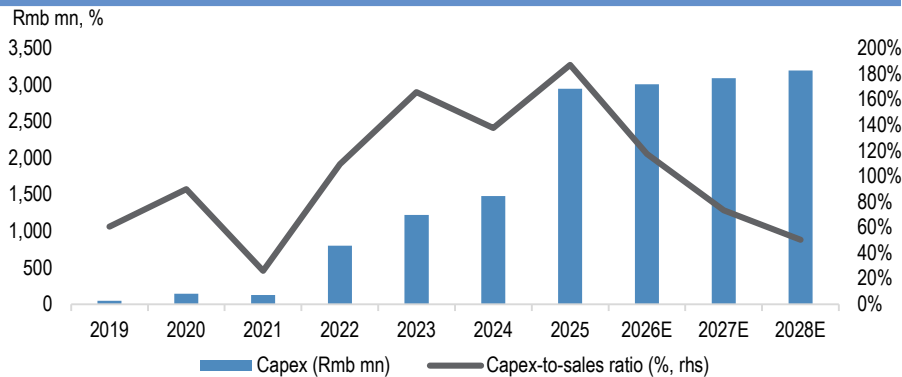
Source: Bloomberg Finance L.P., Wind (6 Aug 2026).

V-Test's favored position to capture AI chip testing demand

□ A counter-cyclical strategy of capacity expansion to cement leadership

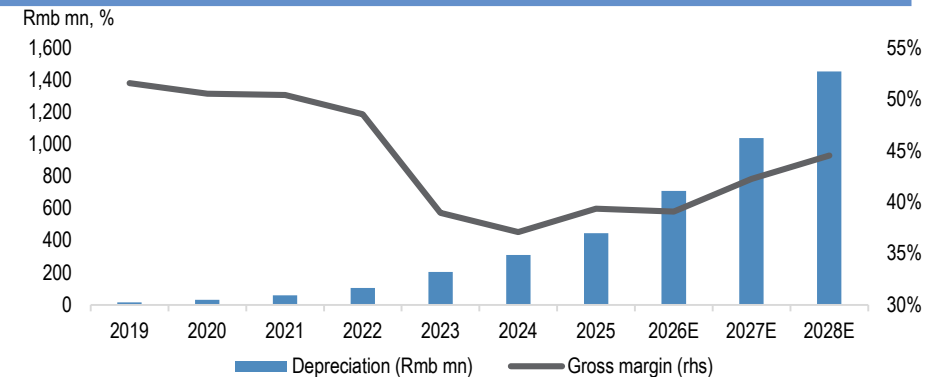
- V-Test's 2025 Capex reached Rmb2.9bn, of which equipment spending was c.Rmb2.5bn, which yields a capex/revenue ratio of 187%. We believe such a capital allocation strategy is commercially rational given high demand certainty and solid customer commitments.
- Although heavy spending had raised concerns over depreciation pressure, capacity utilization stays at high levels (80%+) and depreciation is a structural cost rather than an incremental cost.
 - We model 65%/50% revenue YoY growth into 2027/28E, which outpaces depreciation growth of 46%/40% respectively.
 - Gross margin will naturally recover to 44.6% by 2028E (from a recent trough of 37.1% in 2024) on top of high utilization and a favorable product mix.

V-Test's Capex and Capex-to-sales ratio



Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

V-Test's depreciation and gross margin



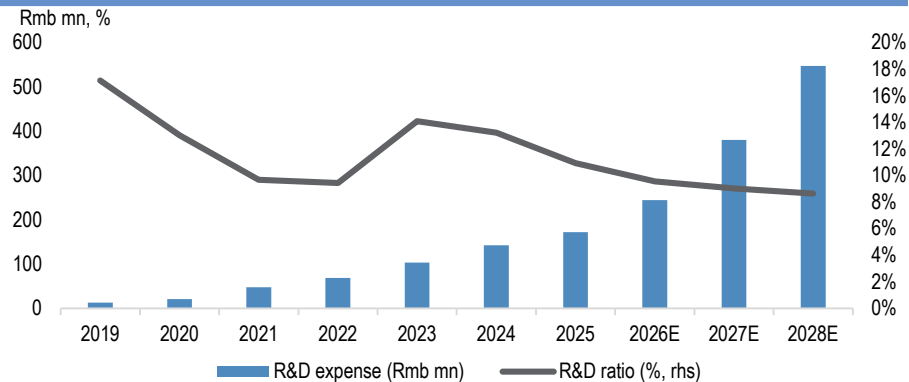
Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

V-Test's favored position to capture AI chip testing demand (cont.)

Moat formation supported by leading equipment fleet and long-lasting R&D investment

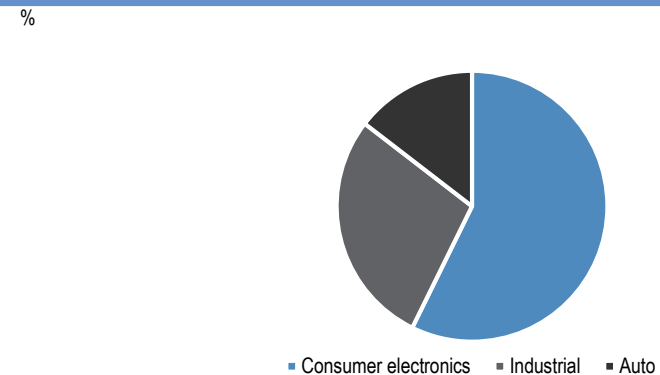
- V-Test is the largest pure-play third-party testing company by installed tester fleet among A-share listed companies. As of end-2025, total testers exceeded 1,500 units, with >400 high-end units.
- Our analysis suggests V-Test's procurement accounted for ~10% of Advantest's 2025 sales in China
- V-Test's R&D expense grew 21% YoY in 2025 (~11% of revenue), demonstrating intensive resources invested for new customers' debut. Against the backdrop of rapidly growing revenue, the structural dilution of the R&D expense ratio is indicative of technology leverage ahead, even though we believe the absolute amount of R&D is still accumulating.
- V-Test covers the vast majority of domestic leading fabless design companies, with its customer footprint spanning consumer SoCs, smartphone SoCs, industrial, automotive, ASAS, AI chips (CPU/GPU/ASIC), and more.
 - By application, consumer electronics contributed 59% in 2025, followed by 29% industrial and 15% automotive.
 - Separating chips by computing power, the sales exposure of computing chips rapidly increased to 20% by end-2025.

V-Test's R&D expense and ratio



Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

V-Test's revenue mix by application (2025)



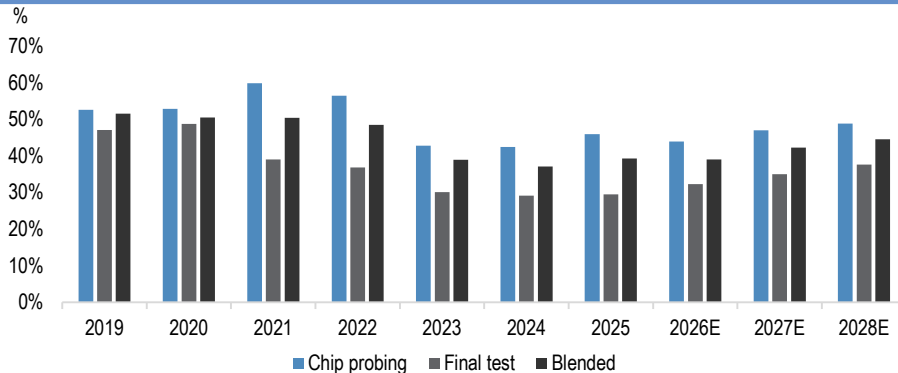
Source: V-Test (2025 annual report).

V-Test's favored position to capture AI chip testing demand (cont.)

□ Profitability optimization with AI chip-driven value enhancement

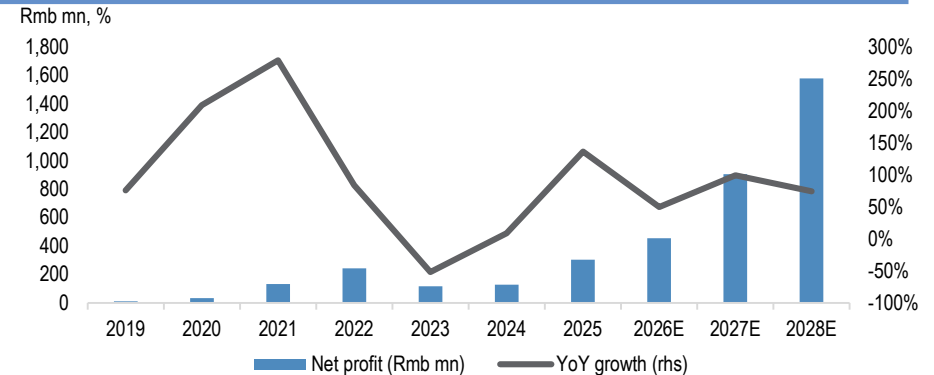
- We forecast V-Test's CP revenue to sustain a ~60% level going forward, modeling a 66% CAGR during 2026-28E (vs 59% for consolidated sales).
 - Domestic foundries systematically underinvested in CP testing.
 - CP business carries a higher gross margin than FT.
- We expect V-Test's compute chip revenue share to continuously rise to >35% by 2028E (from c.20% at end-2025), indicating a ~90% sales CAGR during 2026-28E.
 - AI chip testing commands structurally higher ASPs and margins than consumer SoC testing, largely driven by pricing power created by equipment exclusivity and premium earned by engineering intensity.
- We model 59%/73% sales/earnings CAGR during 2026-28E, thanks to upgrade in both business mix and earnings quality.

V-Test's gross margin trend by segment



Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

V-Test's net profit and YoY growth

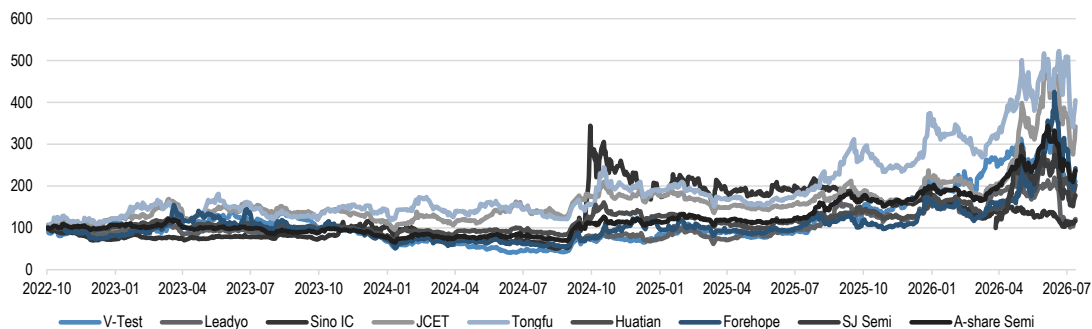


Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

Share performance and valuation

- Since its listing in Oct 2022, V-Test's share price performance has largely tracked the A-share Semiconductor Index (V-Test up 118% vs +145% for the index), while significantly outperforming domestic third-party testing peers thanks to its more rapidly expanding business scale.
- Due to the semi industry cyclical downturn, V-Test and OSAT names underwent a valuation de-rating, from 2H23. A valuation repair began in late 2025; OSAT peers now are trading at roughly one standard deviation above their historical mean, largely supported by pricing elasticity and growing market recognition of their AI chip exposure.
- The structural tailwind of AI demand has propelled Taiwanese peers to record valuation levels, underpinned by rapid earnings growth.
- The stock is currently trading at around 29x one-year forward P/E, representing a 40% discount to the OSAT peer average of 50x one-year forward P/E. We view the current valuation as highly attractive, supported by three key catalysts: 1) multiple expansion driven by increasing AI exposure; 2) an improving ASP trend underpinned by a more favorable product mix; and 3) a more desirable profitability profile relative to OSAT names.

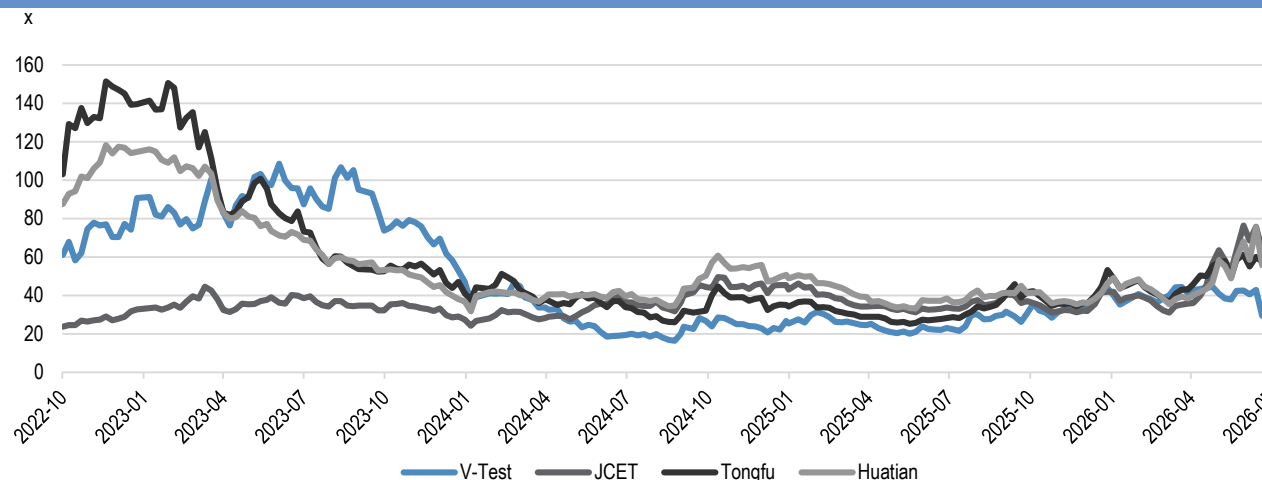
Share price performance comparison – A-share peers



Source: Wind (6 Aug 2026), J.P. Morgan.

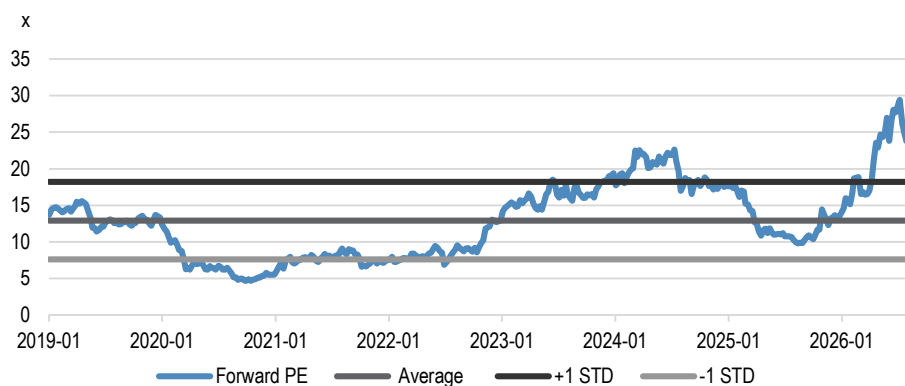
Share performance and valuation (cont.)

Historical forward P/E trend – A-share peers



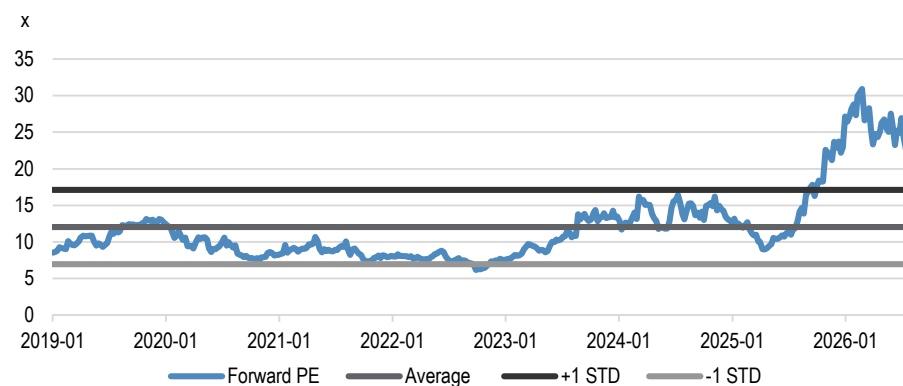
Source: Wind (06 Aug 2026), J.P. Morgan estimates.

ASE's historical forward P/E band



Source: Bloomberg Finance L.P., Wind (06 Aug 2026), J.P. Morgan.

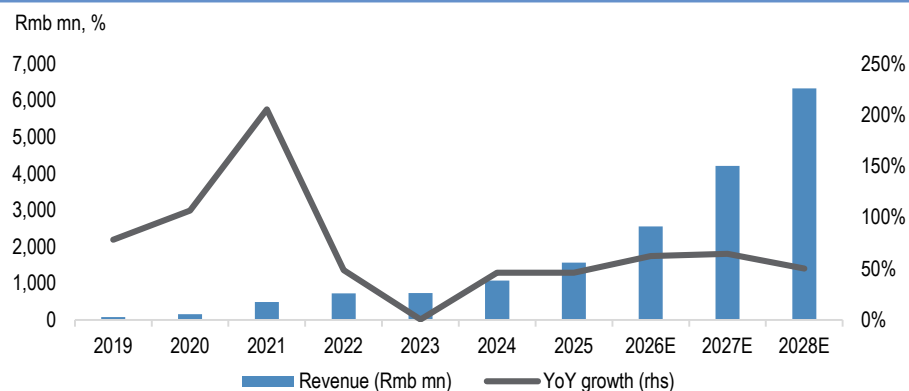
KYEC's historical forward P/E band



Source: Bloomberg Finance L.P., Wind (06 Aug 2026), J.P. Morgan.

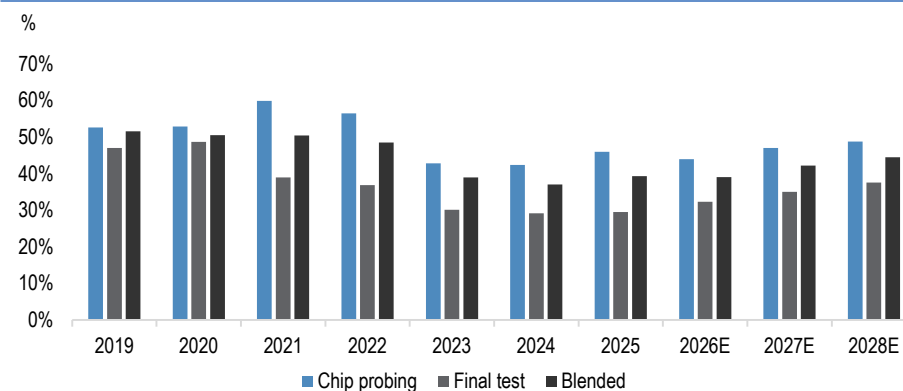
Key financials

V-Test's revenue and YoY growth



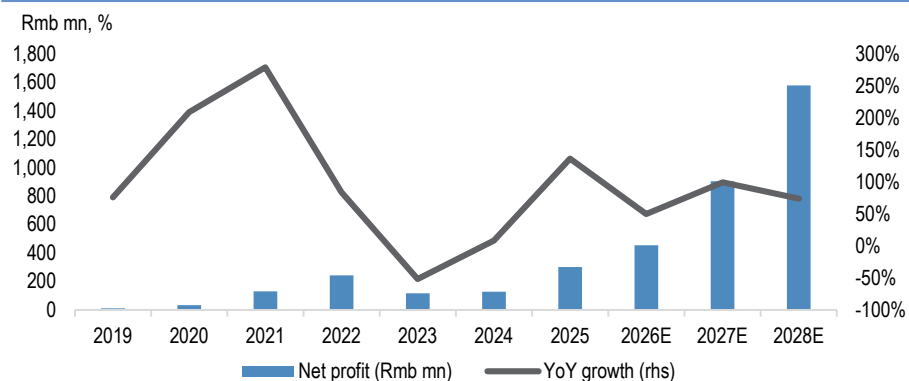
Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

V-Test's margin trend



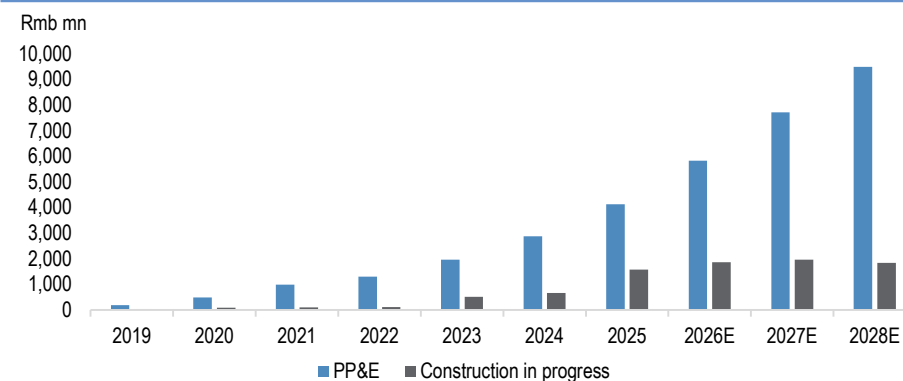
Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

V-Test's net profit and YoY growth



Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

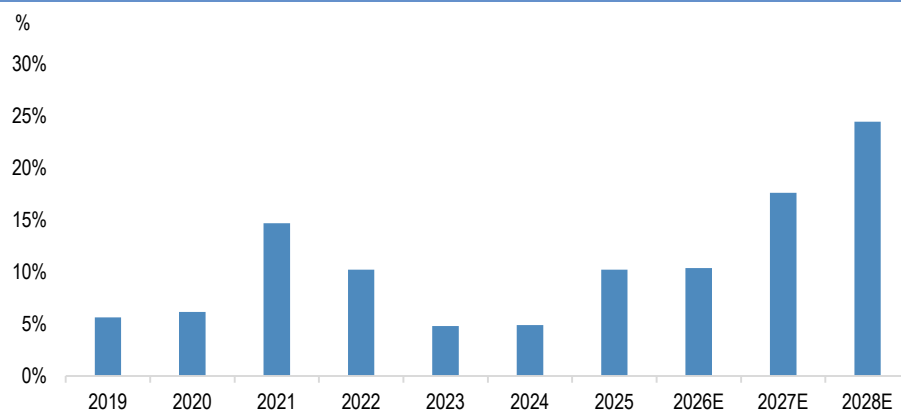
V-Test's PP&E and construction-in-progress trend



Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

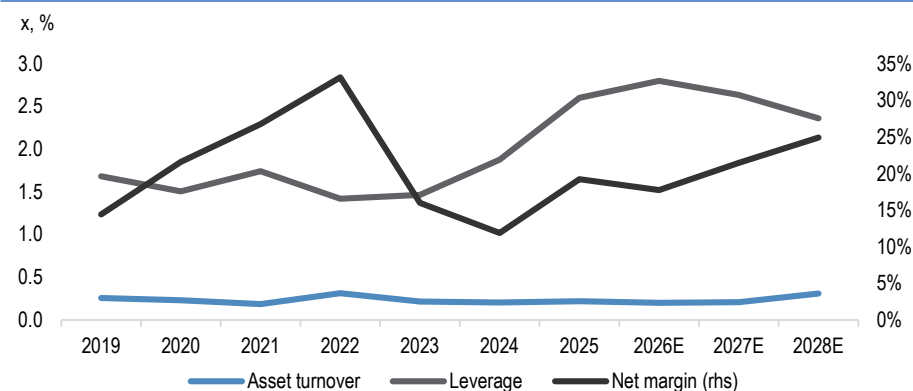
Key financials (cont.)

V-Test's ROE trend



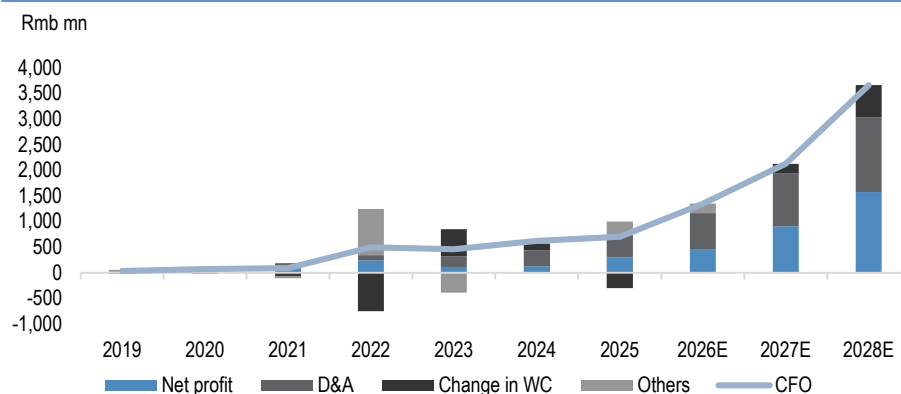
Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

V-Test's DuPont analysis



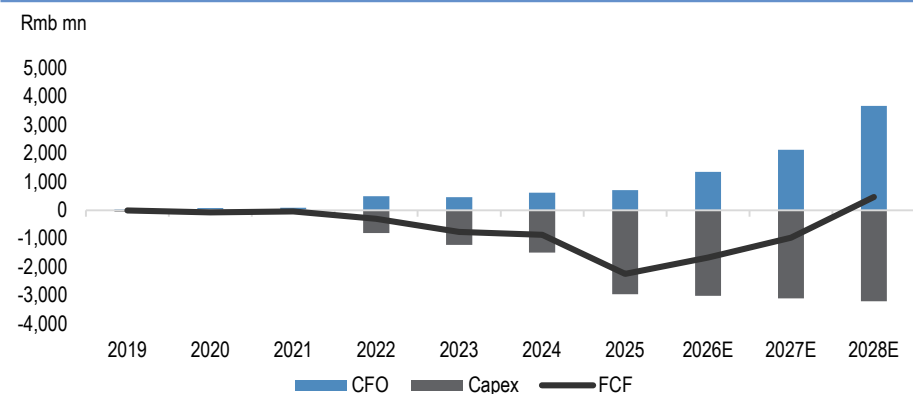
Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

V-Test's operating cash flow trend



Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

V-Test's free cash flow trend



Source: V-Test (Prospectus and historical annual reports, 2019-2025), J.P. Morgan estimates.

Valuation and Investment risks

Valuation

- We derive our Jun-27 target price of Rmb220.00 using a P/E-based approach, applying a 30x one-year forward multiple. Our target multiple benchmarks the company's three-year historical average and sits 30% below the trading average of key A-share OSAT peers, reflecting a discount for its relatively limited pricing flexibility in the near term.

Risks to Rating and Price Target

- Tester procurement restrictions impeding capacity expansion.
- Export controls affecting repatriation of chip testing from foundries outside of mainland China.
- Intensifying industry competition with lower equipment barrier supported by domestic ATE suppliers.
- Weaker-than-expected non-AI demand.

Valuation

V-Test's historical trading band – one-year forward P/E



Source: V-Test (Historical annual reports, 2022-2025), Wind (6 Aug 2026), J.P. Morgan estimates.

V-Test's historical trading band – one-year forward PB/ROE



Source: V-Test (Historical annual reports, 2022-2025), Wind (6 Aug 2026), J.P. Morgan estimates.

Valuation – peer valuation comparison

Ticker	Company	Rating	Market cap (Rmb mn)	YTD perf.	P/E (x)			P/B (x)			ROE			2026-28E Earnings CAGR
					2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E	
688372.SH	V-TEST	OW	20,518	12%	44.8	22.4	12.9	4.6	4.0	3.2	12%	19%	27%	73%
600584.SH	JCET	OW	135,763	107%	60.2	37.2	27.7	4.5	4.1	3.7	8%	11%	14%	46%
688820.SH	*SJSEMI	NC	253,822	78%	220.2	136.8	99.6	12.5	11.5	10.4	6%	8%	10%	40%
002156.SZ	*TFME	NC	92,513	62%	58.5	46.3	40.9	5.5	5.0	4.5	9%	10%	11%	23%
002185.SZ	*HUA TIAN	NC	57,362	58%	57.9	46.6	37.7	3.1	2.9	2.7	6%	6%	7%	29%
688362.SH	*FHEC	NC	29,727	102%	143.2	92.9	63.9	10.4	9.1	7.8	7%	10%	13%	79%
Average					97.4	63.7	47.1	6.8	6.1	5.4	8%	11%	14%	48%

Source: Wind, J.P. Morgan estimates. Note: Prices as of market close on 6 Aug 2026. * Wind consensus for Not Covered (NC) stocks.

Disclosures

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V-Test - A (688372.SH, 688372 CH) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.

The chart(s) show J.P. Morgan's continuing coverage of the stocks; the current analysts may or may not have covered it over the entire period. J.P. Morgan ratings or designations: OW = Overweight, N= Neutral, UW = Underweight, NR = Not Rated

Disclosures

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Coverage Universe: Feng, Billy : AMEC - A (688012.SS), AtHub - A (603881.SS), China Resources Microelectronics - A (688396.SS), Dongshan Precision - A (002384.SZ), Gigadevice Semiconductor - A (603986.SS), Goertek - A (002241.SZ), Hangzhou HikVision Digital Technology Co., Ltd - A (002415.SZ), Huaifeng Test & Control - A (688200.SS), Iluvatar CoreX - H (9903.HK), Inspur - A (000977.SZ), JCET - A (600584.SS), Luxshare - A (002475.SZ), Maxscend Microelectronics - A (300782.SZ), MetaX - A (688802.SH), NAURA - A (002371.SZ), OmniVision - A (603501.SS), Sinnet - A (300383.SZ), Universal Scientific Industrial (Shanghai) - A (601231.SS), Wingtech Tech - A (600745.SS), Zhejiang Dahua Technology Co., Ltd - A (002236.SZ), Zhongji Innolight - A (300308.SZ)

J.P. Morgan Equity Research Ratings Distribution, as of July 04, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
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